A General Theory of Institutional Signal Simplification

Horizontal and Vertical Rolling Windows in Hierarchical Decision Systems

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Abstract

Human institutions do not directly observe reality. Instead, they observe compressed representations generated through layers of aggregation, filtering, summarization, and transmission. This paper develops a General Theory of Institutional Signal Simplification (GTISS) based upon two fundamental mechanisms: horizontal rolling windows and vertical rolling windows. Horizontal rolling windows aggregate information across peer units, while vertical rolling windows aggregate information across hierarchical levels.

The resulting framework unifies signal extraction in governments, corporations, militaries, central banks, scientific organizations, biological systems, financial markets, and artificial intelligence architectures. We establish mathematical foundations, derive information-theoretic results, characterize optimal institutional filtering, introduce an Institutional Nyquist Frequency, and discuss applications across economics, finance, psychology, political science, warfare, geopolitics, machine learning, and complex adaptive systems.

The paper ends with "The End"

Contents

List of Figures

List of Tables

1 Introduction

All institutions face the same fundamental problem.

Reality generates vastly more information than decision-makers can process.

A government cannot observe every economic transaction.

A military commander cannot observe every battlefield event.

A CEO cannot observe every corporate interaction.

A biological organism cannot process every environmental stimulus.

Consequently, all decision systems perform signal simplification.

This paper proposes that institutional information processing can be represented by two coupled filtering mechanisms:

1. Horizontal Rolling Windows.
2. Vertical Rolling Windows.

Together they generate compressed institutional representations of reality.

2 The Institutional Signal Problem

Let

$$X_t \in \mathbb{R}^n$$

denote the true state of the world at time t.

Examples include:

- Economic activity.
- Military deployments.

- Biological conditions.
- Psychological states.
- Political developments.
- Financial transactions.

No institution can fully observe X_t.
Instead, institutions construct

$$S_t = \mathcal{C}(X_t),$$

where

$$\mathcal{C}$$

is a compression operator.

Definition 1. *The institutional signal is the compressed representation*

$$S_t$$

upon which decisions are based.

3 Horizontal Rolling Windows

Suppose there exist

$$N$$

peer units.
Examples include:

- Provinces within a nation.
- Branches of a corporation.
- Military battalions.
- Research laboratories.
- Biological cells.

Let

$$x_{i,t}$$

denote the signal generated by unit i.
Define a horizontal rolling window of length H:

$$\bar{x}_{i,t}^{(H)} = \frac{1}{H} \sum_{k=0}^{H-1} x_{i,t-k}.$$

The institution-wide horizontal aggregate becomes

$$S_t^{(H)} = \frac{1}{N} \sum_{i=1}^{N} \bar{x}_{i,t}^{(H)}.$$

3.1 Interpretation

Horizontal averaging:

- Reduces local noise.
- Preserves common structure.
- Lowers communication costs.
- Improves statistical reliability.

4 Vertical Rolling Windows

Consider a hierarchy with levels

$$l = 0, 1, \ldots, M.$$

Examples include:

Military	Corporate
Soldier	Worker
Sergeant	Supervisor
Captain	Manager
Colonel	Director
General	CEO

Level l receives information from level $l - 1$.
Define

$$S_t^{(l)} = V_l \left(S_{t-W_l+1}^{(l-1)}, \ldots, S_t^{(l-1)} \right).$$

The parameter

$$W_l$$

is the vertical memory horizon.

5 The Double-Rolling Architecture

Reality is represented by

$$X_t = [x_{ij,t}],$$

where

- i indexes horizontal units,
- j indexes hierarchy levels.

Define

$$H_h$$

as the horizontal window length and

$$H_v$$

as the vertical window length.
The institutional signal becomes

$$S_t = \mathcal{V}_{H_v}\Big(\mathcal{H}_{H_h}(X_t)\Big).$$

6 Variance Reduction

Assume

$$x_{i,t} = \mu_t + \epsilon_{i,t},$$

with

$$E[\epsilon_{i,t}] = 0$$

and

$$Var(\epsilon_{i,t}) = \sigma^2.$$

Theorem 1. *If errors are independent, horizontal averaging reduces variance according to*

$$Var(S_t^{(H)}) = \frac{\sigma^2}{NH}.$$

Proof. The variance of an average of NH independent observations equals

$$\frac{\sigma^2}{NH}.$$

□

7 Information-Theoretic Foundations

Let

$$\mathcal{E}(X_t)$$

denote Shannon entropy.
After institutional filtering:

$$\mathcal{E}(S_t).$$

Define the signal simplification ratio

$$\rho = \frac{\mathcal{E}(S_t)}{\mathcal{E}(X_t)}.$$

Definition 2. *The parameter ρ measures retained informational complexity.*

Properties:

$$0 < \rho \leq 1.$$

8 Bandwidth Constraints

Let

$$R$$

denote environmental information generation.
Let

$$B$$

denote institutional communication bandwidth.

Proposition 1. *If*

$$R > B,$$

signal simplification is unavoidable.

Proof. Communication channels cannot transmit information beyond capacity.
Thus aggregation must occur.

□

9 The Signal Simplification Optimization Problem

Institutions solve

$$\min_{H_h, H_v} J(H_h, H_v)$$

where

$$J = \alpha N + \beta L + \gamma C.$$

Here

$$\begin{aligned} N &= \text{Noise}, \\ L &= \text{Lag}, \\ C &= \text{Communication Cost}. \end{aligned}$$

Theorem 2. *Under convexity assumptions, an optimal pair*

$$(H_h^*, H_v^*)$$

exists.

Proof. Assume:

$$J$$

is continuous and coercive.

By the Weierstrass theorem, a minimum exists.

□

10 Institutional Nyquist Frequency

Signal processing suggests that no hierarchy can accurately detect changes occurring above a critical frequency.

Definition 3. *The Institutional Nyquist Frequency is*

$$f_N = \frac{1}{2(H_h + H_v)}.$$

Changes with

$$f > f_N$$

are aliased.

The institution misinterprets reality.

11 Physics Interpretation

The institution resembles a low-pass filter.

Reality generates high-frequency fluctuations.

The hierarchy attenuates them.

The transfer function may be approximated by

$$G(\omega) = \frac{1}{1 + \omega^2 \tau^2}.$$

Large institutions possess larger effective

$$\tau.$$

Consequently:

- Stable.
- Slow.
- Resistant to noise.
- Vulnerable to rapid shocks.

12 Chemistry Interpretation

Chemical reaction networks aggregate microscopic molecular interactions.

Macroscopic observables such as:

- Temperature.
- Pressure.
- Concentration.

are institutional-style summaries of underlying microstates.
Horizontal windows correspond to ensemble averaging.
Vertical windows correspond to thermodynamic coarse-graining.

13 Biological Interpretation

Biological systems exhibit layered signal simplification.

Level	Function
Cells	Detection
Tissues	Aggregation
Organs	Integration
Brain	Decision

Evolution favors filters that maximize survival rather than perfect truth.

14 Psychology and Psychiatry

Human cognition is itself a signal simplification machine.

The brain continuously compresses:

- Visual information.
- Auditory information.
- Social information.
- Emotional information.

Psychiatric dysfunction may arise when filtering parameters become maladaptive.
Examples include:

- Excessive filtering.
- Insufficient filtering.
- Distorted aggregation.

15 Economics

Economic statistics are institutional signals.

Examples:

- GDP.
- Inflation.
- Unemployment.
- Purchasing Managers Index.

Each statistic represents massive compression.
Governments operate on

$$S_t$$

rather than

$$X_t.$$

16 Finance

Financial markets aggregate information.
Market prices become compressed estimators:

$$P_t = E[X_t|\mathcal{F}_t].$$

Institutional investors apply rolling windows through:

- Moving averages.
- Risk models.
- Forecast systems.
- Portfolio construction.

17 Political Science

Political systems contain multiple filtering layers:

1. Citizens.
2. Local officials.
3. Regional officials.
4. National leaders.

Signal distortion increases with hierarchical depth.

18 International Relations

Diplomatic systems function as signal-processing networks.
Information flows through:

- Embassies.
- Intelligence agencies.
- Ministries.
- Heads of state.

Misperceptions emerge from excessive simplification.

19 Welfare Economics

Welfare planners rely on compressed indicators.
The challenge is balancing:

- Administrative simplicity.
- Informational completeness.
- Equity.
- Efficiency.

20 Warfare Economics

Military logistics aggregate information from:

- Fuel inventories.
- Ammunition stocks.
- Personnel counts.
- Supply chains.

Operational success depends upon maintaining signal fidelity.

21 Military Command Systems

The command hierarchy itself is a vertical rolling window.
Information propagates upward.
Orders propagate downward.

Layer	Function
Tactical	Detection
Operational	Aggregation
Strategic	Simplification
Political	Decision

22 Computer Science

Distributed systems perform signal simplification continuously.
Examples include:

- MapReduce.
- Consensus protocols.
- Distributed databases.
- Sensor networks.

23 Data Science

A typical data science pipeline is:

$$\text{Data} \to \text{Features} \to \text{Model} \to \text{Prediction}.$$

This is precisely a signal simplification architecture.

24 Artificial Intelligence

Transformer attention performs adaptive signal aggregation.
State-space models perform rolling summarization.
Institutional architectures resemble deep neural networks:

- Horizontal windows resemble convolution.
- Vertical windows resemble depth.
- Decision outputs resemble classification layers.

25 Algorithmic Representation

25.1 Pseudo-Code

```
Input:
    Raw signals X

For each horizontal unit:
    Compute rolling average

Aggregate horizontally

For each hierarchy level:
    Apply vertical rolling window

Generate institutional signal S

Output:
    Decision D
```

26 Institutional Phase Transitions

Definition 4. *An institutional phase transition occurs when optimal window lengths change discontinuously.*

Examples:

- Financial crises.
- Wars.
- Revolutions.
- Pandemics.

27 Central Theorem

Theorem 3 (Institutional Signal Simplification Theorem)**.** *For an institution with finite communication bandwidth:*

1. *Increasing horizontal window length reduces variance.*
2. *Increasing vertical window length reduces strategic volatility.*
3. *Both increase decision lag.*
4. *There exists an optimal pair*
$$(H_h^*, H_v^*)$$
balancing noise reduction and responsiveness.

Proof. Parts (1) and (2) follow from averaging properties.

Part (3) follows from temporal smoothing.

Part (4) follows from optimization of the objective function

$$J(H_h, H_v).$$

□

28 A Unified Diagram

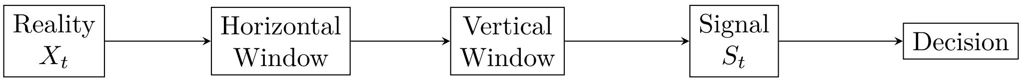

Figure 1: The General Theory of Institutional Signal Simplification.

29 Conclusion

Institutions are fundamentally signal-processing systems.

The General Theory of Institutional Signal Simplification demonstrates that hierarchical organizations can be understood through coupled horizontal and vertical rolling windows operating under bandwidth constraints. The resulting framework unifies concepts from physics, biology, economics, finance, warfare, political science, psychology, computer science, artificial intelligence, and information theory. Institutions therefore do not observe reality directly; rather, they operate upon compressed reconstructions generated through successive layers of filtering. Understanding these filters provides a common language for analyzing organizational performance, crisis response, governance, intelligence systems, and adaptive decision-making in complex environments.

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Glossary

X_t Underlying reality.

S_t Institutional signal.

H_h Horizontal rolling window length.

H_v Vertical rolling window length.

ρ Signal simplification ratio.

B Communication bandwidth.

R Environmental information generation rate.

f_N Institutional Nyquist frequency.

J Institutional optimization objective.

The End